

# Expense Variance & Budget Analysis Report

500-Person Technology Company | January 2023 – December 2024

Includes Power BI Dashboard Insights



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# Executive Summary

This report presents a comprehensive expense variance analysis for a 500-person technology company over 24 months (January 2023 – December 2024). The organisation recorded **55,348 expense transactions** across 8 departments and 29 expense categories. Total actual spend reached **\$159.81M** against a budget of **\$135.57M**, resulting in an overall overspend of **\$24.24M (17.9%)**. Of all transactions, 73% were classified as Overspend, 22% as On-Track, and 5% as Underspend.

|                   |                    |                 |                    |                        |
|-------------------|--------------------|-----------------|--------------------|------------------------|
| <b>\$135.57M</b>  | <b>\$159.81M</b>   | <b>\$24.24M</b> | <b>17.9%</b>       | <b>73%</b>             |
| Total Budget (2Y) | Total Actual Spend | Total Overspend | Overall Variance % | Transactions Overspent |
|                   |                    |                 |                    |                        |

## Key Findings

**Engineering is the largest budget centre.** Engineering accounts for \$75.16M in budget — 55% of total — with a consistent 22.5% overspend pattern across both years.

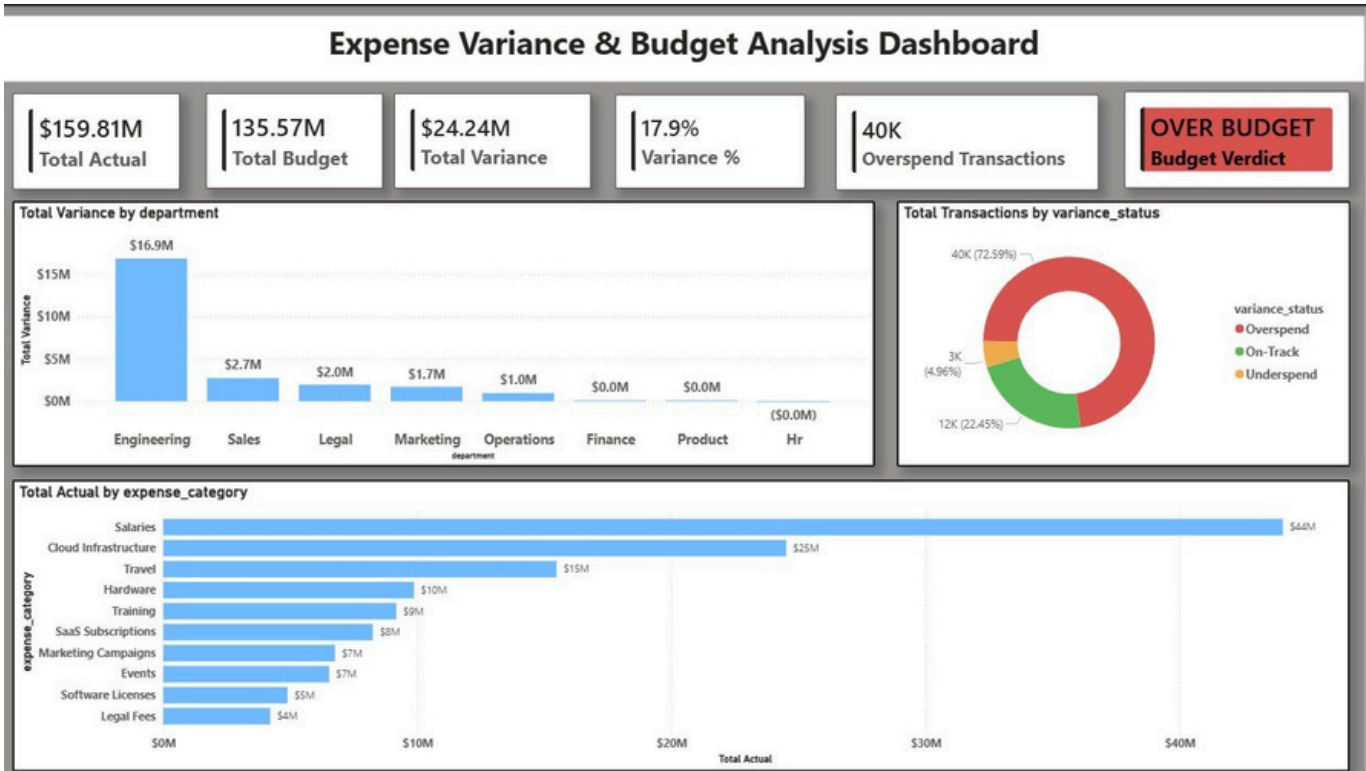
**Three departments are chronic overspenders.** Engineering, Sales, and Legal all exceed budget by more than 22%, driven by salary pressures, travel expenses, and legal fees respectively.

**Marketing and Operations show seasonal patterns.** These departments spike in Q2 and Q4, aligning with campaign cycles and year-end activity, with overspends reaching up to 40% in peak months.

**Finance, HR and Product are well-managed.** These three departments consistently operate within +/-1% of their budgets with forecast accuracy above 99%, serving as benchmarks for budget discipline.

**Salaries and Cloud Infrastructure are the biggest variance drivers.** Salaries contribute \$8.10M and Cloud Infrastructure adds \$4.50M in overspend — together representing over 52% of total variance.

**72.59% overspend rate signals systemic pressure.** Nearly 3 in 4 transactions exceed budget, indicating this is not isolated to anomalies but reflects recurring operational cost pressure across the organisation.



Dashboard Insights — Overview

- Totalactualspendof\$159.81Magainstabudget of \$135.57M represents a \$24.24M (17.9%) overspend, with the "OVER BUDGET" verdict clearly triggered.
- Engineering dominates variance at \$16.9M — more than 6x the next highest department (Sales at \$2.7M), reflecting structural budget misalignment in the largest cost centre.
- 72.59% of all 55K transactions are classified as Overspend (red), with only 22.45% On-Track (green) and 4.96% Underspend (amber).
- Salaries (\$44M actual) and Cloud Infrastructure (\$25M) are the two largest absolute spend categories, together accounting for 43% of total actual spend.
- Travel (\$15M), Hardware (\$10M), and Training (\$9M) round out the top 5, indicating high discretionary spend that may be controllable with tighter policy.

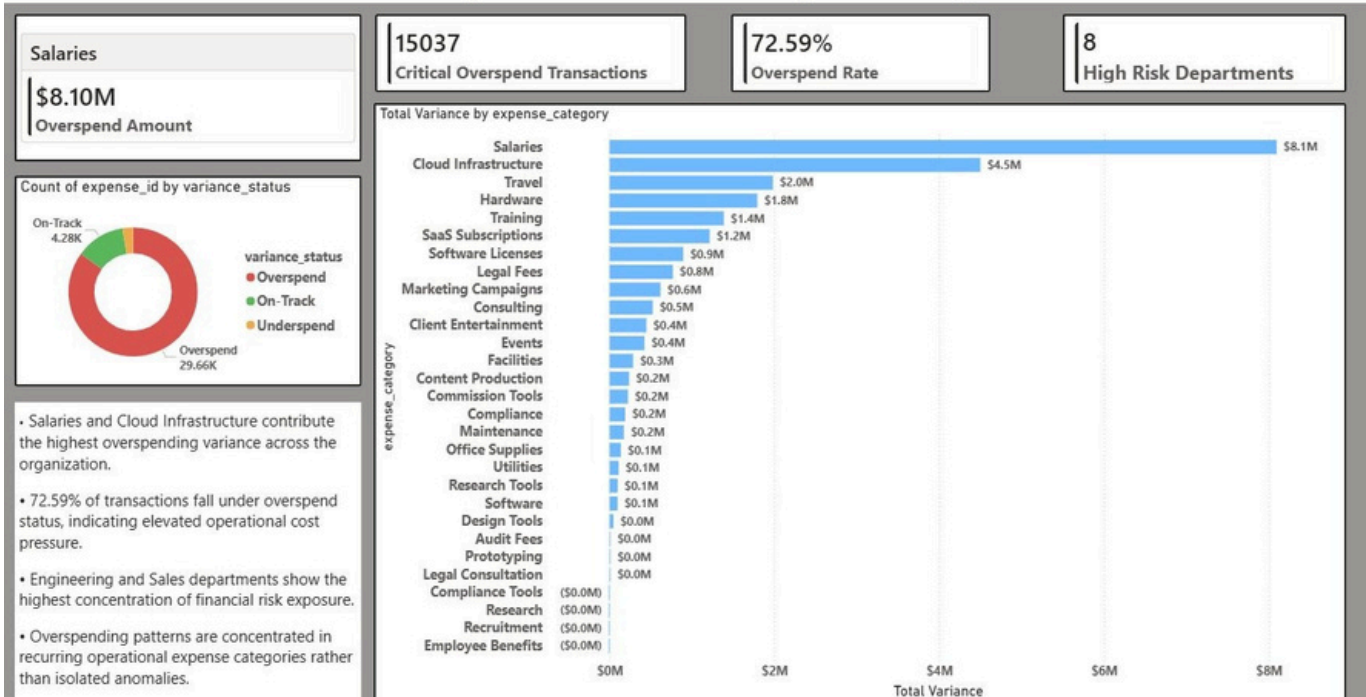
Operational Cost & Department Performance Insights



Dashboard Insights — Department Performance

- Engineering has the highest average spend per employee at \$511.4K — nearly 45% above the company average of \$352.8K — reflecting the cost-intensive nature of engineering infrastructure.
- Legal shows the highest variance % at 22.76%, driven by unpredictable legal fees and consulting engagements that are difficult to forecast accurately.
- Finance and Product achieve near-perfect forecast accuracy of 99.8% and 99.9% respectively, demonstrating that precise budgeting is achievable even at scale.
- HR is the only department with a negative variance (-0.4%), indicating slight underspend, possibly due to unfilled headcount or deferred recruitment activity.
- Overall company forecast accuracy stands at 82.1%, with significant drag from Engineering (77.5%), Sales (77.5%), and Legal (77.2%) pulling the average down.

Expense Risk & Overspend Insights



Dashboard Insights — Risk & Overspend

- 15,037transactions areclassified asCriticalOverspend—representing 27% of all transactions — with Salaries alone contributing \$8.10M in overspend amount.
- All 8 departments are flagged as high-risk, confirming the overspend problem is organisation-wide rather than confined to isolated cost centres.
- The risk donut chart confirms 29.66K overspend transactions vs 4.28K on-track, reinforcing that budget discipline is the exception rather than the norm.
- The long-tail of smaller categories (Compliance Tools, Research, Recruitment, Employee Benefits) show near-zero or negative variance, suggesting well-managed low-volume categories.
- Overspending patterns are concentrated in recurring operational categories — not one-off anomalies — making them structural issues that require policy intervention rather than exception management.

# Department-Level Analysis

Detailed budget vs actual breakdown for all 8 departments over the full 24-month period, ranked by absolute variance.

| Department  | Budget   | Actual   | Variance | Var %  | Status      |
|-------------|----------|----------|----------|--------|-------------|
| Engineering | \$75.16M | \$92.05M | \$16.88M | +22.5% | +22.5% OVER |
| Sales       | \$12.13M | \$14.87M | \$2.73M  | +22.5% | +22.5% OVER |
| Legal       | \$8.62M  | \$10.58M | \$1.96M  | +22.8% | +22.8% OVER |
| Marketing   | \$16.89M | \$18.59M | \$1.71M  | +10.1% | +10.1% OVER |
| Operations  | \$9.71M  | \$10.67M | \$960.2K | +9.9%  | +9.9% OVER  |
| Finance     | \$3.90M  | \$3.91M  | \$8.0K   | +0.2%  | +0.2% OK    |
| Product     | \$4.82M  | \$4.83M  | \$6.5K   | +0.1%  | +0.1% OK    |
| Hr          | \$4.33M  | \$4.32M  | \$-16.1K | -0.4%  | -0.4% OK    |

## Budget Utilisation Bars

Progress bars show actual as % of budget. Red = over 105%, Amber = 100–105%, Green = within budget.

|             |                        |        |           |
|-------------|------------------------|--------|-----------|
| Engineering | <div><div></div></div> | +22.5% | FA: 77.5% |
| Sales       | <div><div></div></div> | +22.5% | FA: 77.5% |
| Legal       | <div><div></div></div> | +22.8% | FA: 77.2% |
| Marketing   | <div><div></div></div> | +10.1% | FA: 89.9% |
| Operations  | <div><div></div></div> | +9.9%  | FA: 90.1% |
| Finance     | <div><div></div></div> | +0.2%  | FA: 99.8% |
| Product     | <div><div></div></div> | +0.1%  | FA: 99.9% |
| Hr          | <div><div></div></div> | -0.4%  | FA: 99.6% |

## Year-over-Year Comparison

| Year | Total Budget | Total Actual | Variance | Variance % |
|------|--------------|--------------|----------|------------|
| 2023 | \$67.69M     | \$79.79M     | \$12.10M | +17.9%     |
| 2024 | \$67.88M     | \$80.02M     | \$12.14M | +17.9%     |

# Expense Category Analysis

Top 15 expense categories by absolute variance, with risk classification.

| #  | Expense Category     | Budget   | Actual   | Variance | Var %  | Risk |
|----|----------------------|----------|----------|----------|--------|------|
| 1  | Salaries             | \$35.97M | \$44.06M | \$8.10M  | +22.5% | HIGH |
| 2  | Cloud Infrastructure | \$20.02M | \$24.52M | \$4.50M  | +22.5% | HIGH |
| 3  | Travel               | \$13.49M | \$15.47M | \$1.98M  | +14.7% | MED  |
| 4  | Hardware             | \$8.08M  | \$9.87M  | \$1.79M  | +22.2% | HIGH |
| 5  | Training             | \$7.79M  | \$9.17M  | \$1.39M  | +17.8% | MED  |
| 6  | SaaS Subscriptions   | \$7.03M  | \$8.25M  | \$1.22M  | +17.3% | MED  |
| 7  | Software Licenses    | \$3.99M  | \$4.89M  | \$894.1K | +22.4% | HIGH |
| 8  | Legal Fees           | \$3.44M  | \$4.21M  | \$768.6K | +22.4% | HIGH |
| 9  | Marketing Campaigns  | \$6.14M  | \$6.76M  | \$618.1K | +10.1% | MED  |
| 10 | Consulting           | \$2.20M  | \$2.72M  | \$524.4K | +23.9% | HIGH |
| 11 | Client Entertainment | \$1.99M  | \$2.44M  | \$447.1K | +22.4% | HIGH |
| 12 | Events               | \$6.10M  | \$6.53M  | \$422.9K | +6.9%  | LOW  |
| 13 | Facilities           | \$2.90M  | \$3.18M  | \$285.6K | +9.9%  | LOW  |
| 14 | Content Production   | \$2.29M  | \$2.53M  | \$237.4K | +10.4% | MED  |
| 15 | Commission Tools     | \$995.9K | \$1.22M  | \$223.5K | +22.4% | HIGH |

## Quarterly Performance

| Year | Quarter | Budget   | Actual   | Variance | Var %  |
|------|---------|----------|----------|----------|--------|
| 2023 | Q1      | \$13.61M | \$15.93M | \$2.32M  | +17.0% |
| 2023 | Q2      | \$18.21M | \$21.69M | \$3.48M  | +19.1% |
| 2023 | Q3      | \$16.13M | \$18.77M | \$2.64M  | +16.4% |
| 2023 | Q4      | \$19.75M | \$23.41M | \$3.66M  | +18.5% |
| 2024 | Q1      | \$13.48M | \$15.79M | \$2.30M  | +17.1% |
| 2024 | Q2      | \$18.35M | \$21.86M | \$3.51M  | +19.1% |
| 2024 | Q3      | \$16.34M | \$19.03M | \$2.69M  | +16.4% |
| 2024 | Q4      | \$19.70M | \$23.34M | \$3.64M  | +18.5% |

# Recommendations

## 1. Implement Hard Budget Caps for Engineering

Engineering's consistent 22.5% overspend on Salaries and Cloud Infrastructure warrants immediate intervention. Monthly budget reviews with the CTO and hard caps in the finance system for cloud spend above \$8,000 per submission are recommended.

## 2. Introduce Pre-Approval for High-Risk Categories

Salaries, Cloud Infrastructure, and Legal Fees collectively account for over 75% of total variance. A mandatory CFO pre-approval workflow for these categories above \$10,000 would significantly reduce overspend leakage.

## 3. Replicate Finance, HR & Product Best Practices

These three departments operate within +/- 1% of budget with >99% forecast accuracy. Their budgeting methodology, monthly forecasting cadence, and approval workflows should be documented and rolled out to Engineering and Sales.

## 4. Seasonal Budget Rebalancing for Marketing & Operations

Given predictable Q2 and Q4 spikes, budgets for these departments should be front-loaded or rebalanced quarterly to avoid technical overspend against flat monthly allocations.

## 5. Automate Variance Alerts at 90% Utilisation

Setup automated alerts when any department or category reaches 90% budget utilisation within a quarter. Early warnings allow managers to course-correct before breaching limits.

## 6. Audit Legal Fees and Consulting Engagements

Legal's 22.8% overspend in Legal Fees and Consulting suggests under-budgeting or scope creep. A formal vendor review and contract renegotiation is recommended for the next fiscal year.

## 7. Address the 72.59% Overspend Transaction Rate

With nearly 3 in 4 transactions exceeding budget, this is a systemic issue. A company-wide budget awareness initiative and monthly department scorecards visible to senior leadership would create accountability and behavioural change.

# Appendix — Data Dictionary

| Field        | Description                                            |
|--------------|--------------------------------------------------------|
| expense_id   | Unique identifier for each expense record (EXP-XXXXXX) |
| employee_id  | Unique employee identifier (EMP-XXXX)                  |
| year / month | Calendar year and month of the expense submission      |
| quarter      | Fiscal quarter (Q1–Q4)                                 |
| department   | One of 8 company departments                           |



| Field                  | Description                                                    |
|------------------------|----------------------------------------------------------------|
| expense_category       | Specific type of expense (29 categories)                       |
| budget_amount          | Monthly budgeted amount for the category/employee              |
| actual_amount          | Actual amount submitted for the expense                        |
| variance_amount        | Actual minus Budget (positive = overspend)                     |
| variance_pct           | Variance as % of budget                                        |
| variance_status        | Overspend (>+5%) / On-Track (±5%) / Underspend (<-5%)          |
| approver               | Approval level based on amount: Manager/Director/VP/CFO        |
| is_outlier             | Boolean flag for statistical outliers                          |
| expense_band           | Spend size band: Low / Medium / High                           |
| variance_severity      | Severity label: On-Track / Moderate / Critical Overspend       |
| budget_utilization_pct | Actual / Budget x 100                                          |
| utilization_category   | High Over Utilized / Over Utilized / On-Track / Under Utilized |
| headcount              | Department headcount used for per-employee calculations        |

### Data Sources & Tools

| Artefact                                 | Description                                                       |
|------------------------------------------|-------------------------------------------------------------------|
| expense_data_cleaned_final.csv           | Primary dataset — 55,348 rows, 22 columns                         |
| generate_dataset.py                      | Python script to generate synthetic 50K+ row dataset              |
| expense_queries.sql                      | SQL queries for database exploration and validation               |
| Expense_Variance___Budget_Analysis.ipynb | Jupyter notebook for EDA and variance analysis                    |
| Dashboard.pbix                           | Power BI file with 3 dashboard pages (Overview, Operations, Risk) |

Report generated on 22 May 2026. All figures in USD. Data covers January 2023 – December 2024 across 8 departments and 29 expense categories.